

Project Atlas — where everything is

A navigation index. It does not re-explain the experiments: the report carries the full record, the defense brief the framing, the cheat sheet the vocabulary. This document answers one question - given a notebook, what does it do and where is it written up; and given a claim, which notebook produced it.

The series letters

letter	series	what it establishes	report home
A	LLM pair (GPT-2 / Pythia)	convergence exists between independently trained language models	Section 3, App A
B	Vision pair (MobileCLIP / SigLIP 2)	it is linear, small, and deployable as a two-tier system	Section 4, App B
C	Unsupervised recovery	the correspondence is partly self-identifying, with no pairs at all	App C.12
E	Cross-modal and geometry	alignment across modalities, its scaling, and what makes it readable	App C.8-C.11, C.14
F	Contrast conditions (training)	what training MORE buys: exactly nothing beyond the linear answer	App C.9, Table C7
G	The shared hub	one coordinate system for five encoders, demonstrated by transfer	App C.13, C.13.1
H	Tooling (not an experiment)	certifies the artifacts every other notebook depends on	App D (method notes)
V	Vector diagnostics	the geometry battery used across the project	App D, cheat sheet

Notebook index

notebook	what it does	headline output	written up in
A1	extract pooled activations, 10k WikiText, 13 layers	activations.npz [13, 10000, 768] per model	Section 3
A2	CKA across all 13x13 layer pairs	the hot diagonal emerges as a RESULT, not a construction	Section 3
A3	linear stitching per layer + controls	R2 0.737-0.798 vs random 0.17	Section 3
B1	extract 4k COCO image pairs, both tiers	pairs.npz (mob_img, sig_img, sig_txt)	Section 4
B2	fit the adapter (closed-form ridge)	W: R2 0.592, cosine 0.900, 0.79 MB	Section 4
B3	retrieval evaluation against the ceiling	R@1/5/10 = 88.4 / 95.1 / 97.2% of ceiling	Section 4
B4	Core ML export for the phone tier	deployable artifact	App B
B6	visual retrieval demo	qualitative check	App B
B7-B9	int8 quantisation: adapter, encoder, vectors	size/quality trade-offs at each stage	App B
B10	vec2vec-style demo on the vision pair	the supervised-baseline comparison for C1	App B
C1	unsupervised matching, first attempt	8.6% exact - epsilon too small, ICP degraded it	App C.12
C1.1	repaired: epsilon swept, ICP dropped, whitening control	39.7% exact = 595x chance ; whitening DESTROYS matching	App C.12
E1	cross-modal fit + the capacity sweep (small/base/large)	40.9 / 53.5 / 58.3% of ceiling, +15.3 pts per decade	App C.8, C.10
E1.1	swap the text encoder to GPT-2	R2 holds (71%), retrieval collapses (34%)	App C.9
E1.2	six label-free corrections on the target space	whitening: R@1 0.130 -> 0.478 = 102.6% of bge	App C.11
E1.3	the encoder ladder, predictions pre-registered	rank-corr +1.00 ; isotropy necessary NOT sufficient	App C.14
E2	cross-modal demo	qualitative check	App C.8
E3	functional stitching at the interface	penalty 0.049	App C
F1	distillation corpus build (38k rows)	teacher-tagged, transfer-checked (0.9168)	App C.9
F2	MLP head distillation - the capacity contrast	0.9052 = 0.9052: converges EXACTLY to linear	App C.9
F3	unfreeze the encoder - built but GATED SHUT	would cost the 'no pretrained weight modified' claim	App C.9
F4	compare and decide	the deploy-gate decision record	App C.9
F5	refit on 38k, evaluate on 5k fresh queries	R@1 +0.0244 real; 90% gate NOT met (89.0%)	App C.9, Table C7
G1	five encoders into one 512-d hub	transfer 95.9% / 92.9% of native on unseen encoders	App C.13
G2	live demo: real images, URL or upload, GPU forward pass	the hub used on novel input	App C.13.1
G3	general N-model compatibility suite + verdict	target-free agreement matrix, gates, controls	(tooling)
H1	power diagnostics: 5 checks before believing a negative	rows/dim, alignment, sanity, target geometry , cache keys	App D method notes
V1	vector geometry battery	effective rank, pair-cosine, hubness	App D.3

Claim to notebook

The reverse lookup: if a number is challenged, this is where it was produced.

claim	number	notebook	section
convergence is linear	MLP = ridge, 0.9052 = 0.9052	F1 -> F2	App C.9
related, not rigid	Procrustes 0.038 vs ridge 0.592	B2	Section 4
deployable	95% of ceiling at 0.79 MB	B2 -> B3	Section 4
decision hierarchy	verif 0.999 / rank 0.987 / member 0.889	B1 + B2 artifacts	App D.2
hubness removed by the adapter	N10 skew 4.28 -> 1.61 (ceiling 1.41)	B1 + B2 artifacts	App D.3
cross-modal without joint training	40.9% of ceiling, 358x chance	E1	App C.8
alignment scales with capacity	+15.3 points per decade, 6 monotone	E1 (three runs)	App C.10
isotropy not similarity training	102.6% / 101.4% at two scales	E1.1 -> E1.2	App C.11
the mechanism predicts	rank-corr +1.00 on two unseen encoders	E1.3	App C.14
self-identifying correspondence	39.7% exact, 595x chance, 0 pairs	C1.1	App C.12
one shared space, used	93-96% component transfer, control at chance	G1	App C.13

Documents

document	what it is for	length
Final_Project_Report.pdf	the full record: body + App A-D	41 pages
Defense_Brief.pdf	assumptions, restrictions, tuning, hard questions, achievements	4 pages
Status_Report.pdf	what has been established, in prose	6 pages
Status_One_Page.pdf	every experiment in one table	1 page
Technical_Cheat_Sheet.pdf	vocabulary, model roster, geometry, the GPT-2 collapse	15 pages
Defense_Deck.pptx	15-minute presentation	13 slides
Project_Atlas.pdf	this document - navigation only	3 pages

Naming note: the notebook formerly called D1 is now H1. Appendix D of the report is the deployment and decision material, so a notebook called D1 pointed at the wrong place. Letters A-G are experiment series; H is tooling that checks measurements rather than producing them.